

Targeted Deep Survival Contrasts: Valid Inference for Individualized Treatment Benefit with Neural Networks

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Abstract

Clinical deployment of neural survival models increasingly requires *predictive* claims—how much a treatment would change survival—not just prognostic risk scores. Such claims demand valid inference for counterfactual survival contrasts under confounding and covariate-dependent censoring, targets for which standard deep survival estimators are biased and provide no honest uncertainty. Building on Targeted Deep Architectures (TDA), which embeds TMLE in a network’s weight space, we target the full vector of treatment-specific survival curves $\{S_1(t), S_0(t)\}$ over a time grid, and hence the benefit curve $S_1(t) - S_0(t)$ and the restricted mean survival time (RMST) difference. A single universal targeting update—a ridge projection of the stacked efficient influence functions onto closed-form last-layer gradients—debiases every coordinate at once, and a multiplier bootstrap yields simultaneous confidence bands for the whole benefit curve. In 200-replication simulations with confounded treatment, sign-varying effect heterogeneity, and censoring that depends on treatment and covariates, our estimator attains nominal pointwise coverage (95.6%) and near-nominal simultaneous band coverage (94.0%), with benefit-curve bias roughly a third of—and intervals 8% narrower than—a per-timepoint one-step (AIPCW) correction built from the same nuisance fits at comparable coverage; a head-to-head mechanism study shows both of TDSC’s ingredients matter, with MSE improving from one-step to output-space survival TMLE to TDSC on identical nuisances, an isotonized one-step closing essentially none of the gap, and causal survival forests carrying visible regularization bias into this population-level target. We provide formal guarantees—joint asymptotic linearity across all coordinates, simultaneous-band validity, and a residual “top-up” that provably restores double robustness—for a cross-fitted variant requiring no Donsker conditions, which is empirically nominal at a quantified efficiency cost (about a third wider intervals). Misspecification and small-sample experiments delineate a three-tier operating envelope: weight-space targeting maximizes efficiency when the initial fit is sound and is the most exposed when it is not, with the output-space TMLE intermediate and the one-step the reliable fallback.

1 Introduction

Neural networks are now standard tools for individual-level survival prediction from rich clinical inputs, including digital pathology and other high-dimensional modalities. Clinical decisions, however, hinge on *contrasts*: how survival would differ under one treatment versus another. Estimating such contrasts from observational data raises two familiar obstacles—confounding and outcome-dependent censoring mechanisms—and a less appreciated third: a network trained for predictive loss is a biased, inefficient estimator of any low-dimensional causal functional [3, 10], and its naive standard errors are invalid.

Semiparametric theory offers the remedy: debias a flexible initial fit by solving the efficient influence function (EIF) equation, via one-step/AIPCW corrections [3, 6] or targeted maximum likelihood estimation (TMLE) [12]. For survival targets these tools are mature [8, 1, 9], but standard implementations correct *outside* the model: the one-step adds the empirical mean of the EIF to a plug-in, and classical TMLE fluctuates the output distribution after training. Both become awkward when the target is a whole *vector*—two survival curves on a fine grid—where per-coordinate corrections need not respect monotonicity and can push estimates off the model space. Targeted Deep Architectures (TDA) [7] instead embeds the TMLE fluctuation *inside* the network: a small targeting subset of weights is updated along the projection of the EIF onto per-sample loss gradients, so the fitted network itself is the debiased estimator. TDA was demonstrated for the average treatment effect and the marginal survival curve without treatment; the clinically decisive object—the causal contrast of treatment-specific survival curves under confounding and dependent censoring—was not addressed.

Related work. Neural estimators of treatment-specific survival curves exist—notably SurvITE [5], which targets exactly our estimand with balanced discrete-time hazard networks, and counterfactual time-to-event models [2]—but they provide point estimates without valid confidence statements; that is the gap we fill. Conversely, survival TMLEs with simultaneous inference exist [1, 9] but target the output distribution through analytic fluctuation submodels that become cumbersome for many coordinates and arbitrary architectures; TDA-style weight-space targeting is what removes that obstruction. Causal survival forests [4] give honest inference for conditional effects at fixed horizons rather than the full population benefit curve.

Contributions. We propose *Targeted Deep Survival Contrasts* (TDSC): (1) the target vector $\Psi = (S_1(t_1), \dots, S_1(t_K), S_0(t_1), \dots, S_0(t_K))$ with its stacked EIFs combining treatment and censoring weights (Section 2); (2) closed-form per-sample last-layer gradients making the universal targeting update for all $2K$ coordinates one ridge regression per iteration (Algorithm 1), with multiplier-bootstrap sup- t bands for the benefit curve (Section 3); (3) formal guarantees—joint asymptotic linearity, band validity, and a residual top-up that provably restores point-estimate double robustness—for a cross-fitted variant satisfying the sample-splitting premise of the underlying ADML theory, together with a design rule (and documented failure mode) for where the high-dimensional targeting step may be solved (Sections 3–4); (4) a comprehensive simulation program—a 200-replication main study, nuisance-misspecification and sample-size grids, a cross-fitting comparison, and head-to-head external comparators (output-space survival TMLE, causal survival forests)—that decomposes the mechanism behind TDSC’s gains and maps its operating envelope (Section 5).

2 Setup, estimands, and efficient influence functions

Data. We observe n i.i.d. copies of $O = (X, A, \tilde{T}, \Delta)$: covariates $X \in \mathbb{R}^d$, binary treatment $A \in \{0, 1\}$, discrete follow-up time $\tilde{T} \in \{1, \dots, K\}$, event indicator Δ . Time is coarsened into K bins; within a bin, events are recorded before censoring, and subjects surviving bin K are administratively censored. Let $Y(k) = \mathbf{1}\{\tilde{T} \geq k\}$ (at risk) and $dN(k) = \mathbf{1}\{\tilde{T} = k, \Delta = 1\}$ (event), and write $P_n f = n^{-1} \sum_i f(O_i)$. The likelihood factorizes through the propensity $g(X) = P(A = 1 | X)$, the event hazard $h(k | a, x)$, and the censoring hazard $h_c(k | a, x)$, with conditional survival $S(t | a, x) = \prod_{k \leq t} \{1 - h(k | a, x)\}$ and censoring survival S_c defined analogously.

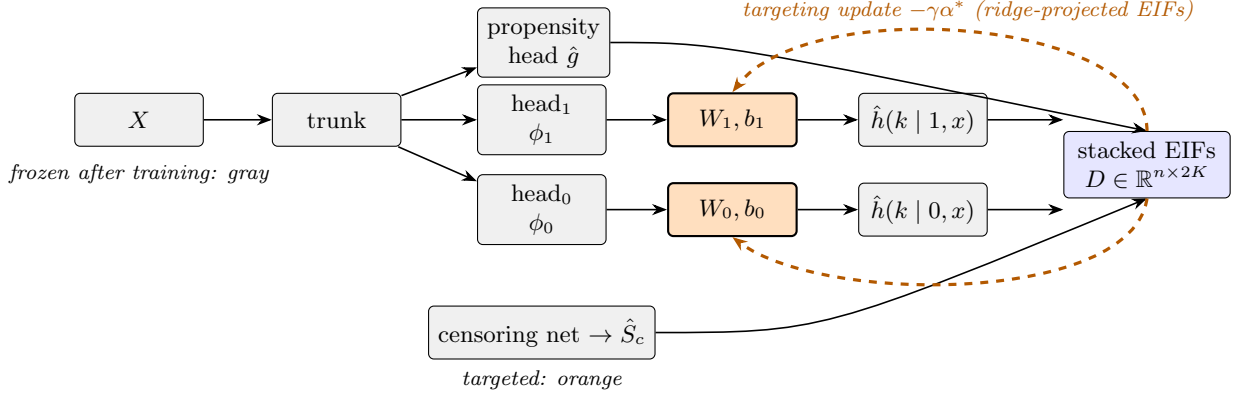


Figure 1: TDSC pipeline. A DragonNet-style survival network produces per-arm hazards; the propensity head and a separate censoring network supply the weights inside the stacked EIFs. Only the final linear layers (W_a, b_a) are updated, along the universal direction obtained by ridge-projecting all $2K$ EIFs onto their closed-form per-sample gradients; everything else stays frozen.

Estimands. Under consistency, no unmeasured confounding, coarsening-at-random censoring (censoring may depend on A and X but not further on the event time), and positivity for treatment and censoring, the treatment-specific curves $S_a(t) = \mathbb{E}_X[S(t | a, X)]$, $a \in \{0, 1\}$, $t \in \{1, \dots, K\}$, are identified. Our target is the $2K$ -vector $\Psi = (S_1(\cdot), S_0(\cdot))$; the *benefit curve* $\beta(t) = S_1(t) - S_0(t)$ and the RMST difference $\Delta_{\text{RMST}} = \sum_{t=1}^K \beta(t)$ (bin width one; the contrast in expected bins survived through horizon K) are linear in Ψ , so their influence functions follow by linearity.

Efficient influence functions. For each coordinate $S_a(t)$ the EIF at P is the discrete-time analogue of Moore and van der Laan [8]:

$$D_{a,t}(O) = - \sum_{k=1}^t \frac{\mathbf{1}\{A=a\}}{g_a(X) S_c(k-1 | a, X)} \frac{S(t | a, X)}{S(k | a, X)} \left(dN(k) - Y(k) h(k | a, X) \right) + S(t | a, X) - S_a(t), \quad (1)$$

with $g_1 = g$, $g_0 = 1 - g$. The first term is a weighted sum of martingale residuals—the within-bin “surprise” of an event relative to the fitted hazard—inversely weighted by the probability of receiving arm a and remaining uncensored through bin $k - 1$; the second is the plug-in centering term. Stacking (1) over a and t gives the $n \times 2K$ influence matrix D whose column means $P_n D_{a,t}$ are exactly the first-order bias terms that targeting drives to zero.

3 Targeted Deep Survival Contrasts

Architecture and initial fit. We use a DragonNet-style network [10] adapted to discrete-time survival: a shared trunk feeding a propensity head for g and two outcome heads producing hazard logits $\{h(k | a, x)\}_{k=1}^K$ for $a \in \{0, 1\}$, trained on the observed-arm discrete-time negative log-likelihood (masked per-bin binary cross-entropy) plus a propensity term, with early stopping. A separate network fits the censoring hazard. Any backbone—including a frozen foundation-model encoder with a trained head—can replace the trunk. Full architecture and hyperparameters are in Appendix A.

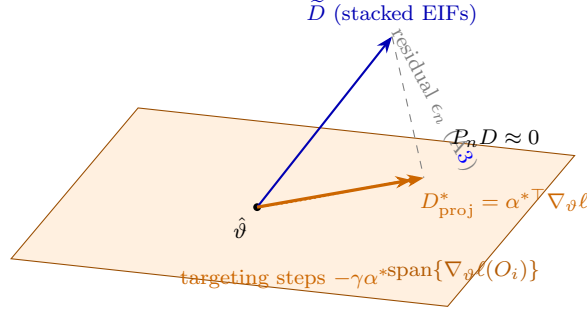


Figure 2: Geometry of the targeting update. The stacked influence functions $\tilde{D}$ are projected onto the span of the per-sample loss gradients of the targeting parameters; TDSC steps ϑ along the projected direction until the empirical EIF equations hold. The projection residual ϵ_n is observable and is exactly the gradient-coverage quantity of Assumption 3: a large residual warns that the last-layer span cannot represent the needed correction (the outcome-misspecified regime of Section 5).

Targeting submodel and closed-form gradients. Following TDA, the targeting parameters ϑ are the final linear layers of the two outcome heads, $p = 2K(F+1)$ parameters for head width F . Because those layers are linear in the head features $\phi_a(X)$ and the loss is a per-bin Bernoulli likelihood, the per-sample gradient is available in closed form:

$$\frac{\partial \ell_i}{\partial W_a[k, \cdot]} = \mathbf{1}\{A_i = a\} Y_i(k) (h(k | a, X_i) - dN_i(k)) \phi_a(X_i), \quad (2)$$

and analogously for biases, giving the $n \times p$ score matrix G in one vectorized pass with no per-sample automatic differentiation.

Universal targeting update. Each iteration solves one ridge system for all $2K$ targets, $\alpha = (G^\top G + \lambda I)^{-1} G^\top D$ (with λ scaled to the mean diagonal of $G^\top G$; note $p > n$ here, so the ridge penalty is statistically active), giving per-target directions α_j and projected bias terms $d_j = P_n[G\alpha_j]$. The universal direction is $\alpha^* = \sum_j (d_j / \|d\|_2) \alpha_j$ [7, Sec. 2.3], applied as $\vartheta \leftarrow \vartheta - \gamma \alpha^*$ with a backtracking line search on $\sum_j (P_n D_j)^2$. After each step the hazards and influence matrix are recomputed (g and S_c stay fixed); iteration stops when every $|P_n D_j| \leq \widehat{\text{sd}}(D_j) / (\sqrt{n} \log n)$ or no step improves the criterion (across our simulations the full tolerance was met in fewer than 1% of replications; otherwise the line search exits first—Section 5 reports the diagnostics). Two diagnostics are recorded: the relative projection residual $\|D - G\alpha\| / \|D\|$ (an empirical check of gradient coverage) and the final $\max_j |P_n D_j| / \text{tol}_j$; any coordinate left above tolerance can receive its residual $P_n D_j$ as a one-step top-up, restoring an exactly solved EIF equation. The estimator is the plug-in from the targeted network—by construction a pair of monotone survival curves.

Cross-fitted variant. To satisfy the sample-splitting premise of the theory below without Donsker conditions on the network class, TDSC admits a V -fold cross-fitted form: for each fold, the nuisance networks are trained *and targeted* on the complement; the held-out fold then supplies evaluation—plug-in curves, influence functions at the targeted fit, and an exact residual correction $P_{n,v}[D]$ so the EIF equations hold exactly on held-out data (by Proposition 3, a cross-fitted one-step at the targeted nuisances). Fold contributions are averaged; the pooled per-observation influence matrix supplies variance and bands.

Algorithm 1 TDSC universal targeting (no-split form)

Require: trained hazard/propensity network and censoring network; targeting parameters ϑ (final layers of the outcome heads); ridge scale λ ; tolerances $\eta_j = \widehat{\text{sd}}(D_j)/(\sqrt{n} \log n)$

- 1: fix weights $\hat{g}, \hat{S}_c$; compute hazards $\hat{h}_a$, survival curves, and stacked EIFs $D \in \mathbb{R}^{n \times 2K}$ via (1)
- 2: **while** $|P_n D_j| > \eta_j$ for some j **do**
- 3: $G \leftarrow$ per-sample loss gradients w.r.t. ϑ $\triangleright$ closed form (2); no autograd
- 4: $\alpha \leftarrow (G^\top G + \lambda I)^{-1} G^\top D$ $\triangleright$ one ridge solve for all $2K$ targets
- 5: $d_j \leftarrow P_n[G\alpha_j]$; $\alpha^* \leftarrow \sum_j (d_j/\|d\|_2) \alpha_j$ $\triangleright$ universal direction
- 6: backtracking line search: $\vartheta \leftarrow \vartheta - \gamma \alpha^*$, accepting only if $\sum_j (P_n D_j)^2$ falls; **break** if no γ improves
- 7: recompute $\hat{h}_a$, curves, and D at the updated ϑ
- 8: **end while**
- 9: **return** plug-ins $\hat{\Psi}_j$; optional residual top-up $+ P_n D_j$ for coordinates above tolerance; SEs $\widehat{\text{sd}}(D_j)/\sqrt{n}$; multiplier-bootstrap band; diagnostics ($\|D - G\alpha\|/\|D\|$, unsolved-coordinate count)

Why not target fold-wise on the held-out folds? A tempting alternative solves the EIF equations with the full targeting submodel *separately on each held-out fold*. This fold-wise high-dimensional targeting fails badly here, and the mechanism is instructive. It is *not* what classical CV-TMLE [11] does: there, a single *low-dimensional* fluctuation is fit by pooling the cross-validated loss over all validation folds— $O(1)$ parameters against n observations—which is exactly why it is safe (the faithful CV-TMLE analogue here would pool the $2K$ one-dimensional fluctuations of Section 5’s output-space TMLE across folds, which we have not run). TDA’s submodel, by contrast, is the network’s final layers ($p \approx 4,000$ here), so on a fold of $n/V \approx 200$ observations the gradient matrix has rank at most 200 and the fold’s empirical EIF equations can be solved nearly exactly—by fitting the fold’s *sampling noise*, whose order $\text{sd}(D)/\sqrt{n/V}$ rivals the genuine first-order bias the step is meant to remove. The resulting hazard perturbations pass through the nonlinear map $h \mapsto \prod_k (1 - h(k))$, rectifying symmetric fold-level fluctuation into systematic bias, and the fold-fitted targeting layers reintroduce exactly the empirical-process term $(P_n - P_0)(\hat{D} - D_0)$ that cross-fitting is meant to eliminate—now concentrated on the smallest split. Empirically this variant gave 14-fold larger benefit-curve bias than the no-split estimator, 70.5% pointwise and 42% simultaneous coverage (100 replications; store preserved in the repository). The design rule: the dimension of the data-adaptive targeting step must be small relative to the split on which its equations are solved, so targeting belongs with training on the large split, and the held-out fold should receive only evaluation and the linear residual correction. Section 5 compares the no-split and cross-fitted forms empirically.

Inference and simultaneous bands. Coordinate standard errors are $\widehat{\text{sd}}(D_j)/\sqrt{n}$ with D evaluated at the final fit; contrasts inherit influence functions by linearity. For the benefit curve we report a sup- t band: with i.i.d. standard normal multipliers ξ_i , the 95% quantile c^* of $\max_t |n^{-1} \sum_i \xi_i D_{\beta,t}(O_i)|/\widehat{\text{se}}_t$ over bootstrap draws gives $\hat{\beta}(t) \pm c^* \widehat{\text{se}}_t$ (c^* floored at 1.96, a conservative choice).

4 Theory

TDSC is TDA applied to a $2K$ -dimensional pathwise-differentiable target, and its guarantees specialize those of Li et al. [7, Thms. 3.1–3.2] and adaptive debiased machine learning (ADML) [13].

We state the specialization for the *cross-fitted* implementation (Section 3; V-fold, nuisances trained and targeted out-of-fold, evaluation with exact residual correction on the held-out fold), which satisfies the sample-splitting premise of those results without Donsker conditions on the network class. Let $M_n \subseteq \mathcal{M}$ be the working submodel induced by partially unfreezing ϑ , $\Psi_n(P_0) = \Psi(\Pi_n P_0)$ the working parameter (projection of P_0 onto M_n), and D_0 the $2K$ -vector of EIFs (1) at P_0 .

Assumption 1 (Identification and positivity). *Consistency, no unmeasured confounding, coarsening-at-random censoring; and $\delta \leq g_0(X) \leq 1 - \delta$, $S_{c,0}(K - 1 \mid a, X) \geq \delta$ a.s. for some $\delta > 0$.*

Assumption 2 (Nuisance rates). *For $f = f(k, a, x)$ write $\|f\|_{P_0}^2 = \sum_a \sum_{k=1}^K \mathbb{E}_{P_0} f(k, a, X)^2$. The cross-fitted weight estimators and the targeted per-fold hazard estimators are each consistent in $\|\cdot\|_{P_0}$, respect the bounds of Assumption 1, and satisfy the product rate $\|\hat{h} - h_0\|_{P_0} (\|\hat{g} - g_0\|_{P_0} + \|\hat{S}_c - S_{c,0}\|_{P_0}) = o_p(n^{-1/2})$ (a sufficient condition for the sharper per- (a, t) sums of per-bin products appearing in the remainder).*

Assumption 3 (Gradient coverage). *The $L^2(P_0)$ -projection of each $D_{0,j}$ onto the span of the final-layer scores $\nabla_{\vartheta} \ell$ has residual $\epsilon_{n,j}$ with $\|\epsilon_{n,j}\|_{P_0} = o_p(1)$ and $(P_n - P_0)\epsilon_{n,j} = o_p(n^{-1/2})$ for each j (under cross-fitting the second clause follows from the first by fold-independence; we state it separately for the no-split reading).*

Assumption 4 (EIF equations solved). $\max_j |P_n D_j| = o_p(n^{-1/2})$ at the final fit (enforced exactly by the residual top-up).

Theorem 1 (Joint asymptotic linearity and efficiency of the top-up estimator). *Under Assumptions 1–4, the cross-fitted TDSC estimator with residual top-up, $\hat{\Psi}^+$, satisfies jointly in $\mathbb{R}^{2K}$ $\hat{\Psi}^+ - \Psi(P_0) = P_n D_0 + o_p(n^{-1/2})$, so $\sqrt{n}(\hat{\Psi}^+ - \Psi(P_0)) \rightsquigarrow N(0, \Sigma_0)$ with $\Sigma_0 = P_0 D_0 D_0^\top$: the estimator is regular and attains the nonparametric efficiency bound. Here Assumption 4 holds by construction (the held-out residual correction solves the EIF equations exactly), and targeting enters through Assumption 2: it must not degrade—and per Assumption 3 is designed to improve—the targeted hazards’ rates.*

Proposition 1 (Cross-fitted plug-in: working parameter and superefficiency). *The cross-fitted TDSC plug-in (no top-up) is instead an ADML-type estimator of the working parameter $\Psi_n(P_0) = \Psi(\Pi_n P_0)$: under the conditions of Li et al. [7, Thms. 3.1–3.2]—including the oracle-bias conditions (C1)–(C2) under which $\Psi_n(P_0)$ may replace $\Psi(P_0)$ —it is asymptotically linear with the working-model EIF, whose variance is no larger than Σ_0 and strictly smaller when P_0 is well-approximated by the submodel (superefficiency). Its empirical signature—lower MSE with conservative coverage relative to Theorem 1’s estimator—appears in Section 5.*

Corollary 1 (Contrasts). *The benefit curve $\hat{\beta}(\cdot)$ and $\hat{\Delta}_{\text{RMST}}$ are asymptotically linear with influence functions $D_{1,t} - D_{0,t}$ and $\sum_t (D_{1,t} - D_{0,t})$, by linearity of both maps.*

Proposition 2 (Simultaneous band). *Under the conditions of Theorem 1, the Gaussian-multiplier estimate of the 95% quantile of $\max_t |\mathbb{G}_t|/\sigma_t$, for $\mathbb{G}$ the limiting Gaussian of Corollary 1, is consistent, and the sup- t band covers the whole benefit curve with asymptotic probability at least 0.95 (the $c^* \geq 1.96$ floor can only enlarge it).*

Proposition 3 (Top-up restores double robustness). *Define the top-up estimator with the full residual, $\hat{\Psi}^+ = \hat{\Psi} + P_n \hat{D}$; it equals the AIPCW one-step evaluated at the targeted nuisances. Consequently, under Assumption 1 and convergence of the estimated nuisances to some limits, $\hat{\Psi}^+$ is consistent for $\Psi(P_0)$ if either the hazard limit equals h_0 or the weight limits equal $(g_0, S_{c,0})$ jointly—regardless of whether targeting converged. (The selective variant implemented in the no-split code, which corrects only coordinates above tolerance, differs from $\hat{\Psi}^+$ coordinate-wise by at most $\eta_j = \widehat{\text{sd}}(D_j)/(\sqrt{n} \log n) = o_p(n^{-1/2})$, so the conclusion is unaffected.)*

Proof sketches are in Appendix B; they specialize the ADML arguments using the exact second-order remainder of the survival EIF (a telescoping identity), with cross-fitting controlling the empirical-process term.

Remark 1 (No-split implementation and scope). Theorem 1 covers the cross-fitted top-up estimator and Proposition 1 the cross-fitted plug-in; the no-split implementation is covered by neither as stated (it violates the sample-splitting premise of TDA’s first-order expansion), and its validity at moderate n is an empirical matter—Section 5 finds nominal behavior at $n \geq 1000$. Two further caveats. Double robustness of the *point estimate* (Proposition 3) does not extend to inference: consistent variance estimation still requires the relevant nuisances, and Section 5 quantifies the failure modes. Superefficiency likewise cuts both ways: narrower intervals when M_n approximates P_0 well, undercoverage when it does not—the empirical price appears under outcome misspecification and at small n .

5 Simulation study

Design. Covariates $X \sim N(0, I_{10})$; treatment $A \mid X \sim \text{Bern}(\text{expit}(0.9X_1 - 0.7X_2 + 0.6X_3))$, truncated to $[0.05, 0.95]$; event hazard $h(k \mid a, x) = \text{expit}(-3.1 + 2k/K + 0.7x_1 + 0.5x_2 + 0.4x_3 - 0.3x_4 + a\tau(x))$ with $\tau(x) = -1 + 0.8x_1 + 0.6x_3$, so the treatment effect changes sign across the population; censoring hazard $\text{expit}(-3.6 + 0.5x_1 + 0.4x_2 + 0.4a + 0.3x_5)$, dependent on treatment and on covariates shared with the outcome (coarsening at random holds; unadjusted estimators are biased). We use $K = 30$, $n = 1000$ ($\approx 61\%$ events, 47% treated in the demo replication). Ground truth is computed by exact integration over 4×10^5 covariate draws; the true Δ_{RMST} is 5.45 bins.

Estimators. (i) **Naive plug-in**: the initial network’s curves; (ii) **unadjusted Kaplan–Meier** by arm; (iii) **one-step (AIPCW)**: per-timepoint $\hat{\Psi}_j^{\text{plug}} + P_n D_j$ at the initial fit, using the same nuisances as TDSC; (iv) **TDSC**, plus its residual top-up variant; (v) in the scenario grid below, an IPTW $\times$ IPCW-weighted KM and an *isotonized* one-step (each arm projected onto monotone curves), which separates how much of TDSC’s advantage is explained by monotonicity alone; (vi) a per-timepoint **output-space survival TMLE**—classical TMLE fluctuating the fitted hazards themselves (“output space,” in contrast to TDSC’s weight-space update), in the style of Moore and van der Laan [8]: each of the $2K$ coordinates gets its own one-dimensional logistic hazard fluctuation with clever covariate $H_t(k, x) = -S(t|a, x)/\{S(k|a, x)g_a(x)S_c(k-1|a, x)\}$, Newton-solved and iterated to convergence, using the *same* nuisance fits as (iii)–(iv); and (vii) **causal survival forests** [4], which fit their own nuisances, evaluated as the doubly robust population benefit at horizons $\{5, 10, \dots, 25\}$.

Results. Table 4 reports time-averaged operating characteristics over 200 Monte Carlo replications (per-cell coverage MC SE ≈ 1.5 points). Four findings. First, adjustment is essential: unadjusted KM’s absolute bias (0.121) is more than 50-fold TDSC’s (0.0021), and its average Δ_{RMST} misses the truth of 5.45 entirely. Second, the naive plug-in carries the regularization-induced bias that debiasing exists to remove (0.021; RMST bias -0.38). Third, on the benefit curve TDSC and the one-step both achieve nominal pointwise coverage (95.6% vs 95.1%), but TDSC does so with roughly a third of the bias (0.0021 vs 0.0057) and 8% narrower intervals (0.140 vs 0.151); the MSE point estimates favor TDSC (0.00149 vs 0.00173), though the *paired* replication-level MSE difference (-0.00024 , MC SE 0.00026) is not resolved at 200 replications. The narrower-at-nominal-coverage combination reflects the working-model efficiency of Section 3; the comparison does not isolate weight-space targeting per se from TMLE-versus-one-step, which the isotonized one-step and output-space TMLE

comparators in the full study are designed to decompose. Fourth, the two estimators trade off differently on the RMST functional: TDSC is essentially unbiased (0.007 vs -0.161) but has higher variance, giving the one-step the lower RMST MSE (0.53 vs 0.79); both cover at $\approx 95\text{--}97\%$. The simultaneous band covered the whole curve in 94.0% of replications, with $c^* \approx 2.9$. Targeting averaged 9.1 iterations. Figure 4 shows one replication (seed 7; representative—per-replication MAE rankings match the table).

Nuisance misspecification and mechanism. A second study (150 replications per arm; propensity fit by a standalone network throughout so the outcome and weight models can be blinded independently) stresses both arms of double robustness by hiding key covariates from one set of nuisances at a time (Table 3). Three conclusions. *Monotonicity is not the mechanism*: in the well-specified arm the isotonized one-step is indistinguishable from the raw one-step (MSE 0.00154 vs 0.00155), while in this configuration TDSC’s paired MSE advantage is clearly resolved (-0.00052 , MC SE 0.00009)—the gain comes from targeting, not from projecting curves onto the monotone cone. *Point-estimate double robustness holds, with a caveat*: blinding the outcome network to the main confounder and effect modifier explodes the naive plug-in (bias 0.19; RMST bias -5.8) while the EIF-corrected estimators stay an order of magnitude less biased; blinding the weight models instead leaves both corrected estimators protected through the hazard arm while the weighting-only IPTW $\times$ IPCW KM fails (bias 0.076).

Inference is not doubly robust, and TDSC pays first: under outcome misspecification TDSC’s working-model intervals undercover badly (64.6%; top-up 84.3%) while the one-step—whose wider intervals track the nonparametric variance—stays at 95.0%. A *safeguarded* variant—the top-up point estimate paired with the nonparametric (initial-fit) SE—repairs much of the shortfall (coverage 88.8%) at the one-step’s width, but only the one-step itself remains fully nominal, consistent with the targeted fit drifting along a badly misspecified gradient span: even the raw TDSC estimate paired with the nonparametric SE covers only 72%, so the failure lives in point-estimate variability that no IF-based SE captures. Under weight misspecification all corrected estimators undercover mildly ($\approx 89\%$) through the variance estimate. Diagnostics flag the failure in-sample: the mean number of coordinates left above tolerance rises from 37 (well-specified; nominal coverage regardless) to 54 (outcome-misspecified), and the strict $\text{sd}/(\sqrt{n} \log n)$ tolerance was never met in full in any of the 450 replications—the line search exits first—so we recommend reporting unsolved-coordinate counts and applying the residual top-up as defaults.

Sample-size scaling. Repeating the main (well-specified, Table 4 configuration) study at $n \in \{500, 2000\}$ (Table 2) shows the expected $\sqrt{n}$ behavior from $n = 1000$ up: TDSC’s $\sqrt{n} \times \text{bias}$ is flat (0.067 at $n=1000$, 0.062 at $n=2000$), $n \times \text{MSE}$ falls (1.49 to 1.02, dipping below the one-step’s 1.43—the superefficiency signature), and pointwise and simultaneous coverage hold at 94–97%. At $n = 500$ the ranking *reverses*: TDSC’s bias (0.0140) exceeds the one-step’s (0.0056) and its coverage drops (91.4% pointwise, 85.3% simultaneous) while the one-step stays near nominal (94.8%). This is the misspecification lesson in another guise—an underpowered initial fit, whether from blinded covariates or a small sample, erodes the working-model gains first—so at small n the one-step is again the safer instrument.

Cross-fitting. The cross-fitted estimator of Sections 3–4 ($V = 5$, 100 replications paired with the main study’s datasets) attains nominal inference everywhere—95.5% pointwise, 94.0% simultaneous, 95.0% RMST coverage—so the estimator covered by Theorem 1 works as advertised. Its price at $n = 1000$: intervals 34% wider than no-split TDSC (0.187 vs 0.140) and higher MSE (paired

difference $+0.00137$, MC SE 0.00022), reflecting nuisances trained on $4n/5$ and the held-out residual correction, whose sampling noise (mean max residual 0.22) more than doubles MSE relative to the cross-fitted plug-in alone (0.00248 vs 0.00104)—the latter the lowest MSE in this comparison, at mildly conservative coverage (97.8%), the empirical signature of Proposition 1. Since both forms are nominal here, the comparison quantifies what the no-split shortcut saves when it is safe: the Donsker concern is empirically immaterial at this scale, and cross-fitting provides the theorem-backed safe harbor—most valuable precisely where no-split inference failed (misspecification, small n), which we leave as the natural next experiment.

External comparators: decomposing the mechanism. Table 1 and Figure 3 report the output-space TMLE and causal survival forests head-to-head with the one-step and TDSC (identical nuisance seeds for (iii), (iv), (vi); 100 replications for the new comparators). In the well-specified arm the three same-nuisance estimators order exactly along the two design axes: one-step (MSE 0.00155) $\rightarrow$ output-space TMLE (0.00127) $\rightarrow$ TDSC (0.00103). Moving from additive correction to TMLE buys roughly half the improvement, and moving the TMLE from output space into weight space buys the other half—so weight-space targeting is not redundant given TMLE-ness, answering the decomposition question the isotonized one-step began. TDSC also has the best coverage of the three (95.7% vs 92.5% for the output-space TMLE, whose per-coordinate fluctuations add small-sample noise). Under *outcome* misspecification the ordering inverts for stability: the output-space TMLE’s one-dimensional fluctuations cannot drift the way the $p \approx 4,000$ -dimensional weight-space update can, so it keeps the lowest MSE (0.00223 vs TDSC’s 0.0186) while only the one-step retains nominal coverage (95.0%)—sharpening the operating envelope: weight-space targeting maximizes efficiency when the initial fit is sound and is the most exposed when it is not, with the output-space TMLE the intermediate-robustness option. Under weights misspecification all three are equivalent (≈ 87 – 89% coverage through the variance estimate). Causal survival forests, solving the harder conditional-effect problem with their own nuisances, carry visible regularization bias into the population benefit (bias 0.018 , roughly $5\times$ the well-specified TDSC bias in Table 1; 89.2% coverage)—all three debiased semiparametric estimators dominate on this population-level target. Finally, the bookkeeping contrast that motivated TDSC: the output-space TMLE required $2K = 60$ separate fluctuation sequences per dataset versus one universal update.

6 Discussion

TDSC turns a two-headed neural survival network into a valid estimator of population treatment benefit at the cost of a few ridge regressions after training, and the misspecification grid gives the method an honest operating envelope: prefer TDSC when the initial network fits well (narrower intervals at nominal coverage, gains not attributable to monotonicity; $n \gtrsim 1000$ in our design), and fall back when diagnostics (unsolved-coordinate counts, projection residuals) suggest otherwise or the sample is small—the safeguarded top-up repairs most of the coverage loss, but under gross outcome misspecification, and at $n = 500$, only the one-step is fully reliable. The remaining work: (i) the cross-fitted variant under misspecification and at small n —it is nominal and modestly less efficient in the well-specified case (Section 5), and the settings where no-split inference failed are exactly where its guarantee should earn its keep; (ii) sensitivity to the ridge penalty, targeting-layer choice, and weight-truncation levels (Appendix A); (iii) *calibration-for-benefit*—targeting arm-specific curves within strata of the network’s own predicted benefit, the estimand used to validate individualized-benefit models against randomized trials—and application on frozen pretrained encoders (e.g. pathology foundation models), which the last-layer construction supports directly.

Table 1: External comparators, benefit curve ($n = 1000$, $K = 30$). One-step and TDSC rows repeat Table 3 (150 replications); output-space TMLE uses the same nuisance seeds (100 replications); causal survival forests fit their own nuisances (100 replications, evaluated at horizons 5–25, so cells are not directly comparable to the 30-point averages). Restricted to the 100 shared replications, the well-specified ordering is unchanged (one-step 0.00158, output-space TMLE 0.00127, TDSC 0.00109).

Scenario	Estimator	Bias	MSE	Coverage	CI width
well-specified	one-step (AIPCW)	0.0051	0.00155	94.5%	0.144
	output-space TMLE	0.0042	0.00127	92.5%	0.134
	TDSC	0.0036	0.00103	95.7%	0.135
outcome mis.	one-step (AIPCW)	0.0135	0.00345	95.0%	0.197
	output-space TMLE	0.0115	0.00223	89.3%	0.149
	TDSC	0.0119	0.01860	64.6%	0.156
weights mis.	one-step (AIPCW)	0.0114	0.00111	89.4%	0.111
	output-space TMLE	0.0132	0.00130	87.1%	0.110
	TDSC	0.0111	0.00114	89.2%	0.111
(own nuisances)	causal survival forest	0.0183	0.00130	89.2%	0.114

Table 2: Sample-size scaling, benefit curve (well-specified, Table 4 configuration; 150/200/150 replications).

n	Estimator	Bias	$n \times \text{MSE}$	Coverage	CI width	Simult. band
500	one-step	0.0056	2.10	94.8%	0.232	—
	TDSC	0.0140	2.77	91.4%	0.198	85.3%
1000	one-step	0.0057	1.73	95.1%	0.151	—
	TDSC	0.0021	1.49	95.6%	0.140	94.0%
2000	one-step	0.0025	1.43	93.7%	0.096	—
	TDSC	0.0014	1.02	96.2%	0.093	96.7%

A Reproducibility

All code, raw Monte Carlo stores, and the exact scripts behind every table are at <https://github.com/blind-contours/tdsc>. Networks: trunk $d \rightarrow 64 \rightarrow 64$ (ELU); propensity head $64 \rightarrow 64 \rightarrow 1$; per-arm outcome heads $64 \rightarrow 64 \rightarrow K$; censoring network $(d+1) \rightarrow 64 \rightarrow 64 \rightarrow (K-1)$. Training: full-batch Adam, lr 10^{-3} , max 500 epochs, early stopping (patience 25) on a 20% validation split, propensity loss weight 1.0. Targeting: ridge scale $\lambda = 0.01 \times \text{mean diag}(G^\top G)$; $\gamma_0 = 1$ with up to 12 halvings; max 50 iterations; tolerance $\hat{\text{sd}}(D_j)/(\sqrt{n} \log n)$. Truncation: $\hat{g} \in [0.05, 0.95]$, $\hat{S}_c \geq 0.05$, conditional survival floored at 10^{-3} inside (1). Bands: 1000 Gaussian-multiplier draws. Cross-fitting: $V = 5$, fold networks trained and targeted on complements with the same hyperparameters. Output-space TMLE: per-coordinate logistic fluctuations solved by Newton steps (ϵ clipped to $[-5, 5]$), recomputing the clever covariate each round, up to 25 rounds per coordinate, same $\text{sd}/(\sqrt{n} \log n)$ tolerance. Causal survival forests: `grf` 2.6.1, `causal_survival_forest` with `target = "survival.probability"`, 1,000 trees, default tuning, one forest per horizon $\{5, 10, 15, 20, 25\}$,

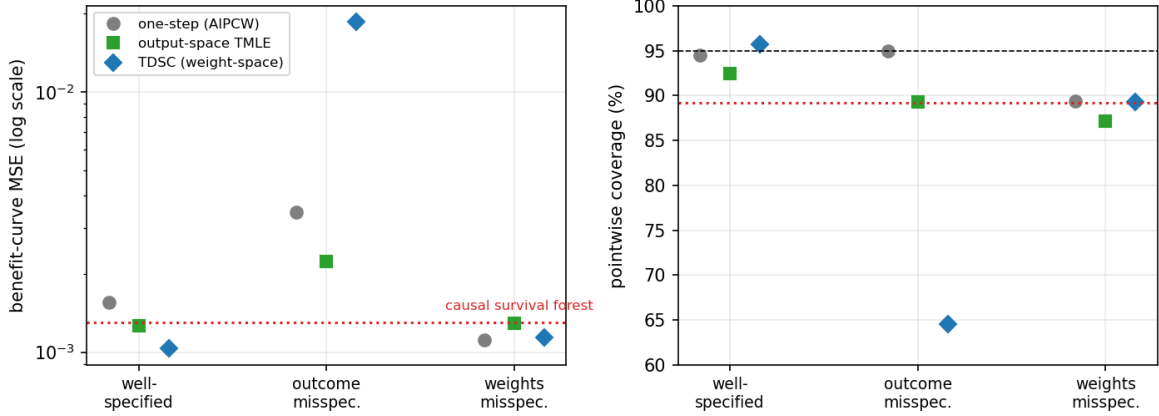


Figure 3: Head-to-head comparison across nuisance scenarios (benefit-curve MSE, log scale; pointwise coverage, 95% dashed). The three same-nuisance estimators order one-step $\rightarrow$ output-space TMLE $\rightarrow$ TDSC when well-specified; under outcome misspecification the weight-space update is the most exposed while the one-step alone stays nominal. The causal survival forest (own nuisances, horizons 5–25) appears as the dotted reference.

population effect and SE from `average_treatment_effect`. The main tables use the no-split estimator (Table 4 with the trunk propensity head; Tables 3 and 1 with a standalone propensity network). Truth: 4×10^5 -draw exact integration.

B Proof sketches

Theorem 1. Per coordinate, since $\hat{\Psi}_j^+ = \Psi_j(\hat{P}) + P_n \hat{D}_j$ and the EIF has the form $\hat{D}_j = \hat{D}_j^{\text{mart}} + \hat{S}_j - \Psi_j(\hat{P})$, the standard one-step decomposition holds exactly:

$$\hat{\Psi}_j^+ - \Psi_j(P_0) = P_n D_{0,j} + (P_n - P_0)(\hat{D}_j - D_{0,j}) + R_{2,j}, \quad R_{2,j} = P_0 \hat{D}_j + \Psi_j(\hat{P}) - \Psi_j(P_0).$$

For the remainder, the survival EIF is exact to second order: writing $S_h(k) = \prod_{j \leq k} (1 - h(j))$ for the conditional survival under a hazard h , the telescoping identity $S_{\hat{h}}(t) - S_{h_0}(t) = \sum_{k \leq t} S_{h_0}(k-1)(h_0(k) - \hat{h}(k)) S_{\hat{h}}(t)/S_{\hat{h}}(k)$ gives $R_{2,j}$ as a sum over $k \leq t$ of integrals of $\{\hat{h}(k) - h_0(k)\}$ against $\{\hat{g} \hat{S}_c(k-1) - g_0 S_{c,0}(k-1)\} / \{\hat{g} \hat{S}_c(k-1)\}$ -type differences; under the positivity bounds of Assumption 1, Cauchy–Schwarz gives $|R_{2,j}| \lesssim \|\hat{h} - h_0\|_{P_0} (\|\hat{g} - g_0\|_{P_0} + \|\hat{S}_c - S_{c,0}\|_{P_0}) = o_p(n^{-1/2})$ by Assumption 2, applied to the targeted per-fold hazards (this display also proves Proposition 3: R_2 vanishes if either factor’s limit is correct). For the empirical-process term, cross-fitting makes $\hat{D}_j$ independent of the fold on which $(P_n - P_0)$ is evaluated, so conditional on the training folds, $(P_n - P_0)(\hat{D}_j - D_{0,j}) = o_p(n^{-1/2})$ whenever $\|\hat{D}_j - D_{0,j}\|_{P_0} = o_p(1)$, which follows from the individual nuisance consistency in Assumption 2 (with Assumption 3 the design mechanism keeping the targeted hazards in the consistent regime). Summing folds yields the coordinate-wise expansion; joint convergence follows from the multivariate CLT applied to the stacked D_0 . Proposition 1 is Li et al. [7, Thms. 3.1–3.2] applied verbatim to the cross-fitted plug-in, with superefficiency from their Theorem 3.2 under (C1)–(C2).

Corollary 1 and Proposition 2. Both maps are bounded linear in Ψ , so asymptotic linearity transfers with the transformed influence functions. Given joint normality with consistently estimated

Table 3: Nuisance-misspecification grid, benefit curve ($n = 1000$, $K = 30$, 150 replications per arm; coverage MC SE $\approx 1.8\text{pp}$). *outcome mis.*: outcome hazards blinded to x_1, x_3 ; *weights mis.*: propensity and censoring models blinded to x_1, x_2 . Propensity is fit by a standalone network (not Table 4’s trunk head), so well-specified cells are not directly comparable to Table 4. The isotonized row changes only the point estimate; the safeguard row changes only the interval (its nonparametric SE equals the one-step’s, hence the equal widths).

Scenario	Estimator	Bias	MSE	Coverage	CI width
well-specified	one-step	0.0051	0.00155	94.5%	0.144
	one-step, isotonized	0.0051	0.00154	—	—
	TDSC	0.0036	0.00103	95.7%	0.135
outcome mis.	naive plug-in	0.1922	0.04110	—	—
	one-step	0.0135	0.00345	95.0%	0.197
	TDSC	0.0119	0.01860	64.6%	0.156
	TDSC + top-up	0.0265	0.01325	84.3%	0.156
	+ nonpar. SE (safeguard)	—	—	88.8%	0.197
weights mis.	IPTW $\times$ IPCW KM	0.0760	0.00720	—	—
	one-step	0.0114	0.00111	89.4%	0.111
	TDSC	0.0111	0.00114	89.2%	0.111

Σ_0 (IF second moments), the conditional multiplier process $n^{-1/2} \sum_i \xi_i \hat{D}_{\beta,t}(O_i)$ converges weakly to the same Gaussian limit (conditional multiplier CLT), so its sup-quantile is consistent for the sup-quantile of the limit; the $c^* \geq 1.96$ floor only enlarges the band.

Proposition 3. By definition $\hat{\Psi}_j^+ = P_n[\hat{S}(t \mid a, X)] + P_n \hat{D}_j$, which is precisely the AIPCW one-step at nuisances $(\hat{h}^{\text{targ}}, \hat{g}, \hat{S}_c)$. Its population bias is the R_2 term above evaluated at those limits, which vanishes under either-arm consistency. $\square$

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Table 4: Benefit curve $S_1(t) - S_0(t)$, time-averaged over 30 points ($n = 1000$, $K = 30$; 200 replications, coverage MC SE $\approx 1.5\text{pp}$). Bias is the time-averaged absolute Monte Carlo bias. The paired TDSC–one-step MSE difference is -0.00024 (MC SE 0.00026).

	Bias	MSE	Coverage (ptwise)	CI width
Naive neural plug-in	0.0213	0.0026	—	—
Unadjusted KM	0.1212	0.0163	—	—
One-step (AIPCW)	0.0057	0.0017	95.1%	0.151
TDSC (ours)	0.0021	0.0015	95.6%	0.140
Δ_{RMST} : one-step	bias -0.161	MSE 0.53	96.5%	2.87
Δ_{RMST} : TDSC	bias 0.007	MSE 0.79	95.5%	2.69
TDSC simultaneous band	coverage 94.0% (whole curve)		band width 0.207	

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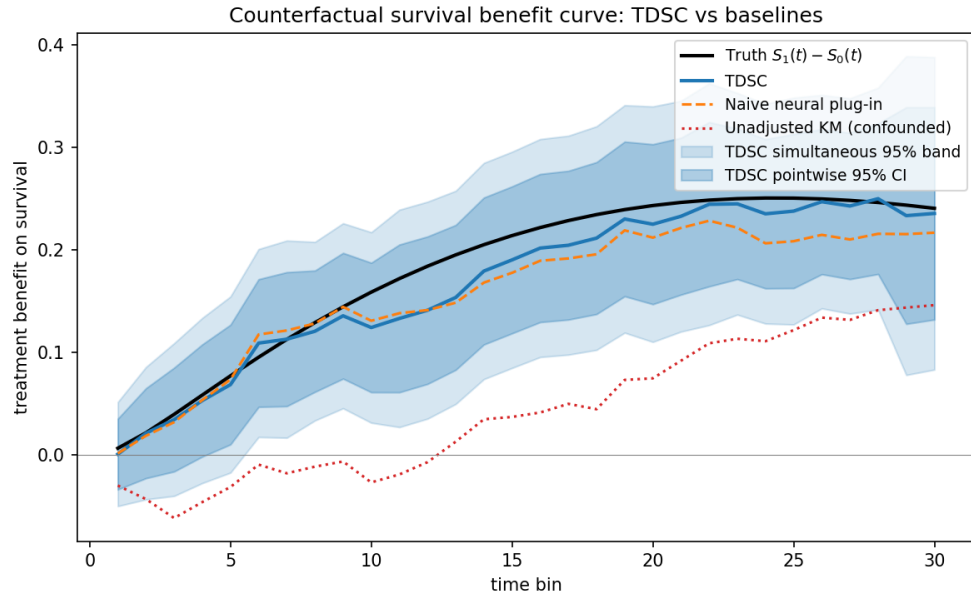


Figure 4: One replication ($n = 1000$, seed 7). The TDSC benefit curve tracks the truth, which lies inside both the pointwise CI and the simultaneous band; the naive plug-in drifts low at later times, and unadjusted KM is severely biased (wrong-signed at early times).